

RESEARCH ARTICLE

Time Delay-Induced Dynamics in Real-Time Hybrid Simulation: Spectral Decomposition and Energy-Based Evaluation

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ABSTRACT

Real-time hybrid simulation (RTHS) inherently functions as a feedback system with intrinsic time delays, which can be accurately modeled using a delay differential equation (DDE). The presence of time delays introduces infinite-dimensional dynamics, complicating the analysis of associated errors. While time delay represents a key experimental imperfection, its quantitative influence on structural vibration remains insufficiently understood. To address this gap, we propose a spectral decomposition framework for linear RTHS systems. This method decomposes the delay system into a finite set of single-degree-of-freedom (SDOF) systems, enabling systematic analysis of delay-induced effects, including frequency shifts, spurious mode generation, and energy redistribution. We establish explicit relationships linking time delay to substructural partitioning and excitation characteristics. Based on these insights, we propose three error mitigation strategies: (1) minimizing actuator delay, (2) reducing the experimental substructure ratio, and (3) optimizing spectral alignment between external excitation and system response. Additionally, we introduce two energy-based evaluation metrics—with corresponding tolerances—to quantify the influence of time delay on total energy input and the modal concentration of input energy. The effectiveness of the proposed approach is validated through numerical simulations and physical experiments, offering novel insights into RTHS error mechanisms from modal and energetic perspectives.

1 | Introduction

As an effective and economical dynamic testing technique, real-time hybrid simulation (RTHS) has attracted widespread attention in structural dynamic analysis and deformation-failure detection of test specimens. Integrated with state-of-the-art dynamic servo-hydraulic actuators, high-performance comput-

ing, communication systems, direct integration algorithms, and compensation/correction methods [1], RTHS facilitates the investigation of rate-dependent behavior in large- or full-scale structures within size limited laboratories [2]. Applications include aerodynamic gust loading alleviation devices in a highly flexible strut-braced wing [3]; and seismic performance of the large-scale space frame structures with high-strength steel composite

Summary

- RTHS can be described as a DDE. This study introduced the spectral decomposition method for projecting the dynamic behavior of DDE to individual modes.
- From the mode and energy perspective, this method can evaluate and quantify how the energy input caused by time-delay is distributed between the inherent and the spurious modes of test system.
- Three ways are effective for error control: (1) reduce the actuator delay, (2) reduce the ratio of experimental substructure, and (3) coordinate spectrums of external excitation and system response.

Y-eccentrically braces [4]. Further applications can be referred to [5–7].

In RTHS, the prototype structure is partitioned into numerical (analytical) and physical (experimental) substructures, where the former is analyzed using the finite element method and the latter is tested in the laboratory. Coupling between computational analysis and experimental loading is maintained through interface compatibility and equilibrium at the interface at each time step. However, seamless data synchronization is challenging due to inevitable response delays, including actuator delay, computation delay, digital-to-analog (D/A) and analog-to-digital (A/D) conversion delays, and communication delay [8]. Actuator delay, which dominates these sources of error, distorts test results and can even damage test equipment. Instability caused by time delay, often leading to unbounded growth in the structural response, is the most common cause of experimental failure [9, 10].

Due to the inherent characteristics of actuator response, RTHS can be interpreted as a feedback system with time delay [11], described by using DDE. Time delay serves as a principal source of experimental error, and understanding its impact is essential to uncovering the fundamental nature of these errors. The relationship between delay error and its impact, is complex and often conflated in existing research, whereas the delay impact cannot be inaccurately judged based solely on delay error. To quantify the time delay, Wang et al. [12], Darby and Williams [13], Ahmadizadeh et al. [2] proposed time-domain evaluation methods, while Guo et al. [14], and Xu et al. [15] introduced frequency domain methods. Hessabi and Mercan [16] combined time- and frequency-domain evaluation in their time-delay evaluation index PAEI. Such approaches focus primarily on quantifying time delay rather than its impact. Indirect methods, such as the tracking indicator [17] and energy errors indicator [18–21] provided partial insights but fail to capture the fundamental impact of time delay on system dynamics.

Stability analysis assesses the macroscopic impact of time delay, only determining whether the dynamic response converges or diverges. Extensive research over the past decade has applied both frequency-domain and time-domain approaches to analyze stability. Wallace et al. [10] investigated the critical stable conditions

of a linear SDOF system based on pole positions. Mercan and Ricles [19, 20] explored the stability in loading-rate-independent and -dependent test specimen, using Routh's stability criterion and the Rekasius polynomial transformation. Chi et al. [21] analyzed delay-dependent stability of neutral RTHS systems using the Padé approximation and root-locus techniques, while Botelho et al. [22] applied the Walton and Marshall method to analyze 1-DOF and 2-DOF single-actuator RTHS systems. By converting a DDE into a generalized eigenvalue problem, Maghareh et al. [23] proposed the predictive stability indicator (PSI) to assess the desynchronization sensitivity at the interface. In contrast, time-domain approaches, such as the Lyapunov-Krasovskii (L-K) theory, can account for time-varying actuator delays. Huang et al. [24–26] demonstrated that the time-varying delays have a more detrimental effect on stability than constant delays. Since RTHS is more appropriately treated as a discrete-time system due to the use of digital devices and numerical integration algorithms, discrete stability analysis has been explored by Chen and Ricles [27], Zhu et al. [28], and Igarashi [29]. Huang et al. [30] further investigated the combined effects of integration algorithms and actuator delays on stability using the discrete L-K theory. Comprehensive insights on stability can be found in [31–33].

Understanding the full modal performance of time delay systems is critical for assessing their dynamic behavior. Wang [34] studied characteristic root locations using an iteration method, while Xu et al. [35] examined boundary and asymptote behavior of characteristic roots. Huang et al. [36] investigated the pole distribution of the time delay system and their influence on stability. These studies reveal that the time delay introduces an infinite number of spurious modes, emphasizing the need for modal analysis in RTHS. However, dynamic behavior analysis of DDE remains limited due to the complexity of transcendental characteristic functions with infinite roots. Functional differential equations [37] is commonly applied to analyze fundamental solutions of time delay systems. Yeniçerioğlu [38] investigated solutions of second-order DDEs with constant coefficients, while Muhsen and Maan [39] applied group analysis and Lie algebra to classify second-order DDE. Bhat and Koivo [40] proposed Luenberger type observers for time-delay systems based on spectral decomposition techniques.

Previous studies provide valuable insights into time delay and its impact, yet they have not fully established the correlation between delay error and delay impact, nor quantified the delay's influence. For example, a high proportion of experimental substructure increases the sensitivity of RTHS to time delay, amplifying experimental distortions even with minor delays. Conversely, a low substructure proportion reduces the overall influence of time delay. Establishing delay control standards for different impacts remains a critical challenge. Analytical solutions for DDE could provide deeper insights, but these solutions are not expressible in elementary functions. To address these limitations, this study introduces a spectral decomposition method that approximates a DDE as a linear superposition of multiple ordinary differential equations (ODE). This approach enhances understanding of time delay dynamics and provides a more accurate framework for quantifying delay impact, guiding effective time-delay control strategies.

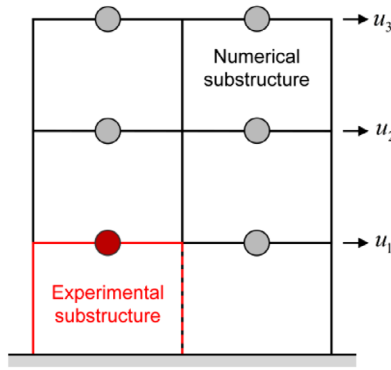


FIGURE 1 | Prototype structure and substructure partition.

2 | Modeling of RTHS

2.1 | Model of RTHS Based on Delay Differential Equation

In RTHS, the prototype structure is partitioned into numerical and experimental substructures, which are synchronously coupled at the substructure interface during each time step. As shown in Figure 1, a typical three-story two-bay planar steel frame [41] serves as the prototype structure. The frame is modeled as linear, with only horizontal shear deformation considered, and lumped masses are located at the middle of each span. The first-story frame on the left side is designated as the experimental substructure (red line), while the remaining frames are assigned for the numerical substructure (black line).

Assuming that the above model has n degrees-of-freedom (DOFs), the n -order matrices $\mathbf{M}^N$, $\mathbf{C}^N$ and $\mathbf{K}^N$ represents the mass, damping and stiffness of numerical substructure, respectively; the n -dimension vectors $\mathbf{u}(t)$, $\dot{\mathbf{u}}(t)$ and $\ddot{\mathbf{u}}(t)$ are the structural displacement, velocity and acceleration responses in the global coordinate system (GCS); the $n \times 1$ vector $\mathbf{F}^E(t)$ is the measured restoring force of experimental substructure; and t represents time. When the system is subjected to an external force $\mathbf{f}(t)$, the equation of motion (EOM) governing the response of RTHS is given as follow.

$$\mathbf{M}^N \ddot{\mathbf{u}}(t) + \mathbf{C}^N \dot{\mathbf{u}}(t) + \mathbf{K}^N \mathbf{u}(t) + \mathbf{F}^E(t) = \mathbf{f}(t) \quad (1)$$

Define u^{cmd} as the command (target) displacement for the experimental substructure; $\mathbf{\Gamma}$ as the $1 \times n$ geometric transformation matrix [42], which transforms $\mathbf{u}(t)$ in GCS to $u^{\text{cmd}}(t)$ in the local coordinate system (LCS). For example, the command displacement in Figure 1 is $u^{\text{cmd}}(t) = u_1(t)$, $\mathbf{\Gamma} = [1 \ 0 \ 0]$.

$$u^{\text{cmd}}(t) = \mathbf{\Gamma} \mathbf{u}(t) \quad (2)$$

Assuming that the mass, damping and stiffness of test specimen are m^E , c^E , and k^E , respectively, the restoring force $f^{\text{mea}}(t)$ in LCS can be expressed as follow, where the measured displacement $u^{\text{mea}}(t)$ represents the actual loading displacement of actuator.

$$f^{\text{mea}}(t) = m^E \ddot{u}^{\text{mea}}(t) + c^E \dot{u}^{\text{mea}}(t) + k^E u^{\text{mea}}(t) \quad (3)$$

In ideal conditions, actuator response is both timely and accurate, that is, $u^{\text{mea}} = u^{\text{cmd}}$. However, in practice, servo-hydraulic dynamics introduce tracking errors between u^{mea} and u^{cmd} . To address this, researchers have developed various actuator response models, such as the servo-hydraulic dynamic model [43], the constant delay model [9], the time-varying delay model [17], the discrete transfer function model [20], etc. Among these, the constant delay model, which assumes a fixed delay actuator delay τ between u^{mea} and u^{cmd} , is the simplest and most computationally efficient approach.

$$u^{\text{mea}}(t) = u^{\text{cmd}}(t - \tau) \quad (4)$$

Transforming f^{mea} in LCS back to $\mathbf{F}^E$ in GCS,

$$\mathbf{F}^E(t) = \mathbf{\Gamma}^T f^{\text{mea}}(t) \quad (5)$$

Setting $\mathbf{M}^E = \mathbf{\Gamma}^T m^E \mathbf{\Gamma}$, $\mathbf{C}^E = \mathbf{\Gamma}^T c^E \mathbf{\Gamma}$, $\mathbf{K}^E = \mathbf{\Gamma}^T k^E \mathbf{\Gamma}$ and substituting Equations (2)–(5) into Equation (1), the EOM of RTHS can be expressed as follow.

$$\begin{aligned} \mathbf{M}^N \ddot{\mathbf{u}}(t) + \mathbf{C}^N \dot{\mathbf{u}}(t) + \mathbf{K}^N \mathbf{u}(t) + \mathbf{M}^E \ddot{\mathbf{u}}(t - \tau) + \mathbf{C}^E \dot{\mathbf{u}}(t - \tau) \\ + \mathbf{K}^E \mathbf{u}(t - \tau) = \mathbf{f}(t) \end{aligned} \quad (6a)$$

Equation (6a) is a delay differential equation (DDE), demonstrating that RTHS inherently behaves as a time-delay system. Under ideal conditions where actuator response is timely and accurate, Equation (6a) reduces to an ordinary differential Equation (6b).

$$(\mathbf{M}^N + \mathbf{M}^E) \ddot{\mathbf{u}}(t) + (\mathbf{C}^N + \mathbf{C}^E) \dot{\mathbf{u}}(t) + (\mathbf{K}^N + \mathbf{K}^E) \mathbf{u}(t) = \mathbf{f}(t) \quad (6b)$$

Currently, the dynamic behavior governed by ODE has been extensively studied by Clough and Penzien [44] and Chopra [45], these theories cannot be easily generalized to DDE.

2.2 | State-Space Formulation and Transfer Function

Equation (6a) describes a linear time-invariant (LTI) system, which can be expressed by the state-space formulation as

(7a). The $2n$ -dimension state vector $\mathbf{x}(t) = \begin{bmatrix} \mathbf{u}(t) \\ \dot{\mathbf{u}}(t) \end{bmatrix}$; the

$$2n\text{-order state matrix } \mathbf{A}_0 = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -(\mathbf{M}^N)^{-1} \mathbf{K}^N & -(\mathbf{M}^N)^{-1} \mathbf{C}^N \end{bmatrix},$$

$$\mathbf{A}_1 = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ -(\mathbf{M}^N)^{-1} \mathbf{K}^E & -(\mathbf{M}^N)^{-1} \mathbf{C}^E \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -(\mathbf{M}^N)^{-1} \mathbf{M}^E \end{bmatrix};$$

the input matrix $\mathbf{B} = \begin{bmatrix} \mathbf{0} \\ (\mathbf{M}^N)^{-1} \end{bmatrix}$; the initial state of system is static, $\Phi(t) = \mathbf{0}$ when $t \in [-\tau, 0)$.

$$\dot{\mathbf{x}}(t) - \mathbf{A}_2 \dot{\mathbf{x}}(t - \tau) = \mathbf{A}_0 \mathbf{x}(t) + \mathbf{A}_1 \mathbf{x}(t - \tau) + \mathbf{B} \mathbf{f}(t) \quad (7a)$$

Moreover, according to functional differential equation theory, when the mass of the experimental substructure is significantly smaller than that of the numerical substructure, $(\mathbf{M}^N)^{-1} \mathbf{M}^E \rightarrow \mathbf{0}$, Equation (7a) is categorized as a retarded functional differential equation (RFDE); otherwise, it is a neutral functional differential

equation (NFDE). The primary difference between RFDE and NFDE lies in whether the highest-order derivative contains a delay term, making RFDE is a special case of NFDE with $\mathbf{A}_2 = \mathbf{0}$. Applying the Laplace transform to Equation (7a) leads to the transfer function in Equation (7b).

$$\mathbf{H}(s) = \mathbf{X}(s) \mathbf{F}^{-1}(s) = (s\mathbf{I} - \mathbf{A}_0 - \mathbf{A}_1 e^{-\tau s} - \mathbf{A}_2 s e^{-\tau s})^{-1} \mathbf{B} \quad (7b)$$

Under ideal condition ($\tau = 0$), the state-space formulation can be expressed as (8a), and its transfer function can be written as Equation (8b), where

$$\mathbf{a} = \begin{bmatrix} \mathbf{0} \\ -(\mathbf{M}^N + \mathbf{M}^E)^{-1}(\mathbf{K}^N + \mathbf{K}^E) & -(\mathbf{M}^N + \mathbf{M}^E)^{-1}(\mathbf{C}^N + \mathbf{C}^E) \end{bmatrix},$$

$$\mathbf{b} = \begin{bmatrix} \mathbf{0} \\ (\mathbf{M}^N + \mathbf{M}^E)^{-1} \end{bmatrix}.$$

$$\dot{\mathbf{x}}(t) = \mathbf{a}\mathbf{x}(t) + \mathbf{b}\mathbf{f}(t) \quad (8a)$$

$$\mathbf{h}(s) = \mathbf{X}(s) \mathbf{F}^{-1}(s) = (s\mathbf{I} - \mathbf{a})^{-1} \mathbf{b} \quad (8b)$$

Since Equation (7b) includes the exponential delay term $e^{-\tau s}$, analyzing the gain and phase margins becomes challenging. To address this, we aim to simplify $\mathbf{H}(s)$ to a polynomial form, similar to $\mathbf{h}(s)$ in Equation (8b), such polynomial representation suggests that the DDE can be decomposed to multiple ODEs. This study focuses on achieving this simplification by using the theory of functional differential equations [37].

3 | Spectral Decomposition of NFDE

The most effective approach to understanding the dynamic behavior of DDE and quantifying the impact of time delay is through analytical solutions. However, DDE solutions cannot be expressed by elementary functions. This section will introduce the spectral decomposition method to analyze the dynamic behavior of time delay system.

For the purpose of analysis and illustration of the proposed method, we focus on the simplest RTHS case, that is, a SDOF test system with a constant delay ($n = 1, \Gamma = 1$). Accordingly, $\mathbf{u}, \mathbf{f}, \mathbf{M}^N, \mathbf{C}^N, \mathbf{K}^N, \mathbf{M}^E, \mathbf{C}^E, \mathbf{K}^E$ reduce to $u, f, M^N, C^N, K^N, M^E, C^E, K^E$, respectively.

3.1 | Principle of Spectral Decomposition

Based on the principle of modal equivalence, spectral decomposition maps the dynamic behavior of DDEs onto individual modal components, thereby reducing the infinite-dimensional time delay system to a finite superposition of single-mode systems. This method facilitates the evaluation of energy redistribution between inherent- and spurious- modes.

For the state-space formulation (7a), we want to build an equivalent state-space formulation as follow.

$$\frac{d}{dt} x_t = A x_t, x_0 = \eta \quad (9)$$

Then, approximate the time delay system to a superposition of single-mode systems, which includes three specific steps.

Step 1: Spectral decomposition

The time delay system has infinite-order modes, such modes together form a generalized eigenspaces C . Solve the initial n -order eigenvalues $(\lambda, \bar{\lambda})$ and eigenvectors $(\phi, \bar{\phi})$, which satisfy Equations (10a) and (10b).

$$\det(\lambda\mathbf{I} - \mathbf{A}_0 - (\mathbf{A}_1 + \mathbf{A}_2\lambda)e^{-\lambda\tau}) = 0 \quad (10a)$$

$$(\lambda\mathbf{I} - \mathbf{A}_0 - (\mathbf{A}_1 + \mathbf{A}_2\lambda)e^{-\lambda\tau})\phi = \mathbf{0} \quad (10b)$$

The initial j -order modes form a subspace P_Λ , the else forms subspace Q_Λ , generalized eigenspaces is composed by P_Λ and Q_Λ , $C = P_\Lambda \oplus Q_\Lambda$, the symbol $\oplus$ means the direct sum.

Step 2: Modal projection

After constructing a subspace P_Λ composed of the first n -modes, project A in Equation (9) onto this subspace. The projection operator $E_P : A \rightarrow P_\Lambda$ can be characterized by Equation (11), where projection method is shown in Appendix.

$$E_P(x) = \Phi_\Lambda \langle \Psi_\Lambda, x \rangle \quad (11)$$

Step 3: Equivalence substitution

Performing the projection operator E_P on both sides of Equation (9) leads to

$$x_t^P(t) = \Phi_\Lambda(0) \langle \Psi_\Lambda, x_t \rangle = \Phi_\Lambda y^P(t) \quad (12a)$$

$$y^P(t) = A_\Lambda y^P(t) + \Psi_\Lambda(0) B f(t), y^P(0) = \langle \Psi_\Lambda, \eta \rangle \quad (12b)$$

Applying the Laplace transform to Equations (12a)–(12b) leads to the approximate transfer function $\tilde{\mathbf{H}}(s)$.

$$\tilde{\mathbf{H}}(s) = \mathbf{X}^P(s) \mathbf{F}^{-1}(s) = \Phi_\Lambda(0) (s\mathbf{I} - \mathbf{A}_\Lambda)^{-1} \Psi_\Lambda(0) \mathbf{B} \quad (12c)$$

Through the above three steps, the infinite-modes time delay system Equation (7a) is reduced to a superposition of finite single-mode systems in Equation (12b), which facilitates the evaluation of modal energy distribution.

3.2 | Delay Impact and Energy Redistribution

Compared the ideal transfer function (8b) to the main mode transfer function $\tilde{\mathbf{H}}^1(s)$ in Equation (12c), the main mode excursion caused by time delay can be expressed as Equation (13a), where Ω and Ω_1 indicates the original mode of $\mathbf{h}(s)$ and the main mode of $\tilde{\mathbf{H}}^1(s)$, respectively. The magnitude variation of main mode can be written as Equation (13b).

$$\rho_{\text{freq}} = \frac{\Omega_1}{\Omega} \quad (13a)$$

$$\rho_{\text{mag}}(\omega) = \frac{|\tilde{H}^1(i\omega)|}{|h(i\omega)|} \quad (13b)$$

Equations (13a)–(13b) represent the delay impacts of main mode excursion and magnitude variation. Furthermore, we can evaluate the impact of time delay from energy perspective. Integrating both sides of Equation (6a) along with the motion path leads to Equation (14a), where the right side represents the total energy input E from the external force. For the ideal test system, the total energy input E_0 can be written as Equation (14b).

$$\int_C (\mathbf{M}^N \ddot{\mathbf{u}} + \mathbf{C}^N \dot{\mathbf{u}} + \mathbf{K}^N \mathbf{u} + \mathbf{M}^E \ddot{\mathbf{u}}(t - \tau) + \mathbf{C}^E \dot{\mathbf{u}}(t - \tau) + \mathbf{K}^E \mathbf{u}(t - \tau)) d\mathbf{u} = \int_C \mathbf{f} d\mathbf{u} = E \quad (14a)$$

$$\begin{aligned} \int_C ((\mathbf{M}^N + \mathbf{M}^E) \ddot{\mathbf{u}} + (\mathbf{C}^N + \mathbf{C}^E) \dot{\mathbf{u}} + (\mathbf{K}^N + \mathbf{K}^E) \mathbf{u}) d\mathbf{u} \\ = \int_C \mathbf{f} d\mathbf{u} = E_0 \end{aligned} \quad (14b)$$

Wallace [10] pointed out that time delay leads to an increase in energy input of RTHS. Define ρ_{Eg} as the total energy input between time delay system and ideal test system, $\rho_{\text{Eg}} = 1$ means the time delay does not cause additional energy input.

$$\rho_{\text{Eg}} = \frac{E}{E_0} \quad (15)$$

According to Equation (12c), the motion path of time-delay system can be approximated to Equation (16), where n represents the approximated order.

$$\mathbf{u}(s) \approx \tilde{\mathbf{H}}^n(s) \mathbf{F}(s) = \sum_{j=1}^n \tilde{\mathbf{u}}^j(s) \quad (16)$$

Hence, the energy input of system in the j^{th} -mode is E^j , and its proportion in the total energy E is denoted as ρ_E^j .

$$E^j = \int_C \mathbf{f} d\tilde{\mathbf{u}}^j \quad (17a)$$

$$\rho_E^j = \frac{E^j}{E} \quad (17b)$$

The above equations decompose the total energy input from external excitation into individual modes. This decomposition is crucial for assessing whether the external excitation introduces additional energy into the RTHS system via spurious modes induced by time delay.

3.3 | Energy Assessment

Equations (15) and (17b) propose two energy based evaluation indexes, ρ_{Eg} reflects the variation of energy input caused by

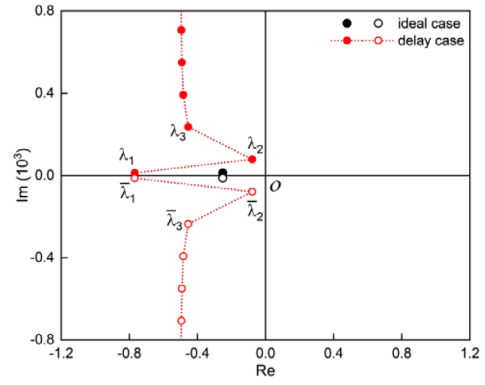


FIGURE 2 | Characteristic roots distribution.

time delay, ρ_E^1 reflects the concentration of energy input in the main mode. Equations (18a) and (18b) provides the tolerance of such indexes, implying that when the energy input deviation of RTHS is less than 5%, meanwhile, more than 99% of energy input is concentrated in the main mode, we acknowledge the test error is under controlled and the test results are reliable.

$$0.95 \leq \rho_{\text{Eg}} \leq 1.05 \quad (18a)$$

$$\rho_E^1 \geq 0.99 \quad (18b)$$

4 | Numerical Analysis

Consider a SDOF structure, of which mass is 1 kg, damping is 0.5027 N.s/m and stiffness is 157.914 N/m. The numerical and experimental substructures are $M^N = 0.505$ kg, $M^E = 0.495$ kg, $C^N = 0.3770$ N.s/m, $C^E = 0.1257$ N.s/m, $K^N = 105.276$ N/m and $K^E = 52.638$ N/m. The actuator delay τ is 0 and 0.04 s for ideal and delay case, respectively. The initial state is static for $t \in [-\tau, 0)$.

4.1 | Spectral Analysis

According to Equations (7b) and (8b), the transfer functions of the ideal and time-delay system are

$$\mathbf{h}(s) = \frac{1}{(M^N + M^E)s^2 + (C^N + C^E)s + K^N + K^E} \begin{bmatrix} 1 \\ s \end{bmatrix}$$

and $\mathbf{H}(s) = \frac{1}{M^N s^2 + C^N s + K^N + (M^E s^2 + C^E s + K^E)e^{-\tau s}} \begin{bmatrix} 1 \\ s \end{bmatrix}$, respectively. Moreover, based on Equations (10a)–(10b), Table 1 and Figure 2 show the eigenvalue and eigenvector of the time-delay system. It is observed that such system has infinite pairs of eigenvalues, $(\lambda_1, \tilde{\lambda}_1) \dots (\lambda_\infty, \tilde{\lambda}_\infty)$ and $\text{Im}(\lambda_\infty) \rightarrow \infty$, indicating the presence of an infinite number of modes. The following section shows the first-, second-, and third-order approximation, where a finite number of modes are used to approximate the infinite modes.

Note $A_\Lambda^j = \begin{bmatrix} \text{Re}(\lambda_j) & \text{Im}(\lambda_j) \\ -\text{Im}(\lambda_j) & \text{Re}(\lambda_j) \end{bmatrix}$. In the first-mode approximation, $A_\Lambda = A_\Lambda^1$,

TABLE 1 | Eigenvalue and Eigenvector of the time-delay system.

Mode	Ideal Case (1st)	Delay Case (1st)	Delay Case (2nd)	Delay Case (3rd)
Frq (Hz)	2	1.99	12.53	37.51
$\lambda_j, \bar{\lambda}_j$	$-0.2513 \pm 12.564i$	$-0.7690 \pm 12.511i$	$-0.07795 \pm 78.702i$	$-0.4539 \pm 235.67i$
$\phi_j, \bar{\phi}_j$	$\begin{bmatrix} 0.0016 \mp 0.0793i \\ 0.9968 \end{bmatrix}$	$\begin{bmatrix} -0.488 \times 10^{-2} \mp 0.0794i \\ 0.9968 \end{bmatrix}$	$\begin{bmatrix} -1.258 \times 10^{-5} \mp 0.0127i \\ 0.9999 \end{bmatrix}$	$\begin{bmatrix} -8.173 \times 10^{-6} \mp 0.424 \times 10^{-2}i \\ 1 \end{bmatrix}$

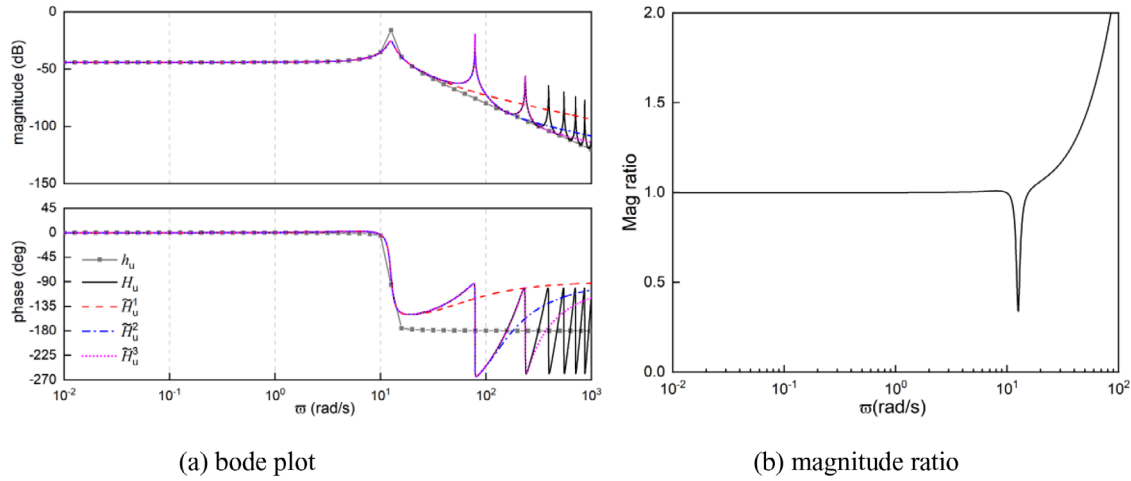


FIGURE 3 | Displacement bode plot of ideal test system, time-delay system, and approximated system.

$$\Phi(\theta) = e^{-0.7690\theta} \begin{bmatrix} 0.1123\sin(12.511\theta) - 0.69 \times 10^{-2}\cos(12.511\theta) & -0.1123\cos(12.511\theta) - 0.69 \times 10^{-2}\sin(12.511\theta) \\ 1.4097\cos(12.511\theta) & 1.4097\sin(12.511\theta) \end{bmatrix}$$

$$\Psi(\xi) = e^{0.7690\xi} \begin{bmatrix} 8.860\sin(12.511\xi) - 0.7517\cos(12.511\xi) & 0.3451\cos(12.511\xi) + 0.1123\sin(12.511\xi) \\ -8.860\cos(12.511\xi) - 0.7517\sin(12.511\xi) & 0.3451\sin(12.511\xi) - 0.1123\cos(12.511\xi) \end{bmatrix}$$

The approximated transfer function $\hat{\mathbf{H}}^1(s) = \begin{bmatrix} \hat{H}_u^1(s) \\ \hat{H}_v^1(s) \end{bmatrix} = \frac{1}{s^2 + 1.538s + 157.1219} \begin{bmatrix} 0.0203s + 0.9944 \\ 0.9633s - 3.1829 \end{bmatrix}$. Similarly, in the second-mode approximation, $A_\Lambda = \begin{bmatrix} A_\Lambda^1 & 0 \\ 0 & A_\Lambda^2 \end{bmatrix}$, so $\hat{\mathbf{H}}^2(s) = \begin{bmatrix} \hat{H}_u^2(s) \\ \hat{H}_v^2(s) \end{bmatrix} = \hat{\mathbf{H}}^1(s) + \frac{1}{s^2 + 0.1559s + 6194.08} \begin{bmatrix} -0.0165s + 0.0231 \\ 0.0256s + 102.2 \end{bmatrix}$. In the third-mode approximation, $A_\Lambda = \begin{bmatrix} A_\Lambda^1 & 0 & 0 \\ 0 & A_\Lambda^2 & 0 \\ 0 & 0 & A_\Lambda^3 \end{bmatrix}$, so $\hat{\mathbf{H}}^3(s) = \begin{bmatrix} \hat{H}_u^3(s) \\ \hat{H}_v^3(s) \end{bmatrix} = \hat{\mathbf{H}}^2(s) + \frac{1}{s^2 + 0.9079s + 55541.4} \begin{bmatrix} -0.1789 \times 10^{-2}s - 0.9364 \times 10^{-3} \\ 0.6877 \times 10^{-3}s + 99.361 \end{bmatrix}$.

Figure 3a shows the Bode plots for the ideal test system, the time-delay system, and the approximated systems for displacement response. The grey dot line and black lines represent the frequency response functions of ideal and time-delay systems, respectively. The ideal system exhibits a single mode at 2 Hz,

while the time-delay system has infinite modes, with the first-, second-, and third-modes at 1.99, 12.53, and 37.51 Hz, respectively. Compared to the ideal system, the main (first) mode of time-delay system experiences a slight excursion of -0.42% . Figure 3b show that the magnitude ratio at the peak of main mode reduces to 33.5%, and the error in the main mode's magnitude ratio is under 5% at 1.76 Hz.

To better understand the spectral characteristics of the time-delay system, $\mathbf{H}(s)$ can be decomposed into individual modes. The red, blue and pink lines represent the first-, second-, and third-approximated frequency response function $\hat{\mathbf{H}}^1(s)$, $\hat{\mathbf{H}}^2(s)$, and $\hat{\mathbf{H}}^3(s)$, respectively. $\hat{\mathbf{H}}^n(s)$ approaches to $\mathbf{H}(s)$ as the order n increases. Figure 4a shows the magnitude ratio $\frac{|\hat{H}_u^n(i\omega)|}{|H_u(i\omega)|}$, where the approximation error for the first-, second-, and third-order models remains below 5% for frequencies less than 33.3 rad/s (5.3 Hz), 126 rad/s (20.05 Hz), and 254 rad/s (40.4 Hz), respectively. Figure 4b illustrates the phase error, which satisfies the 5%

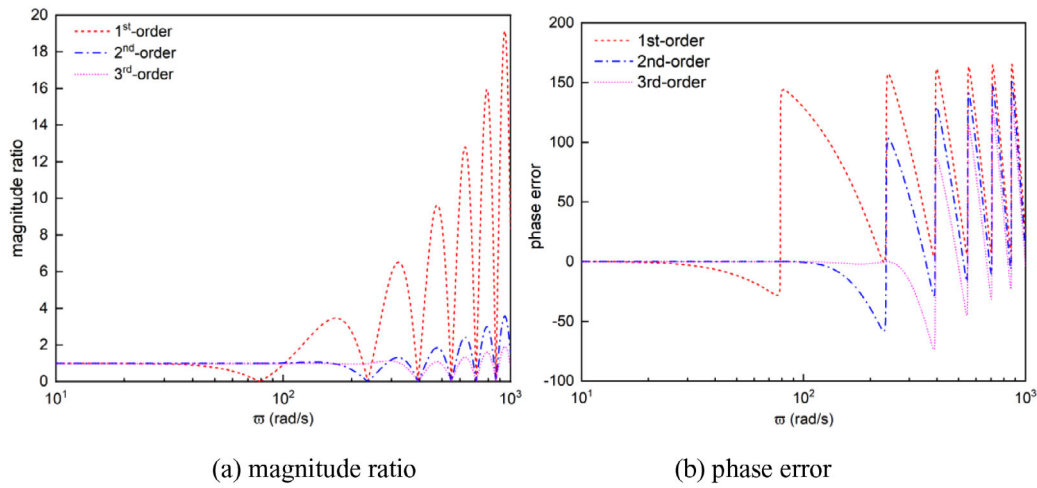


FIGURE 4 | Magnitude ratio and phase error.

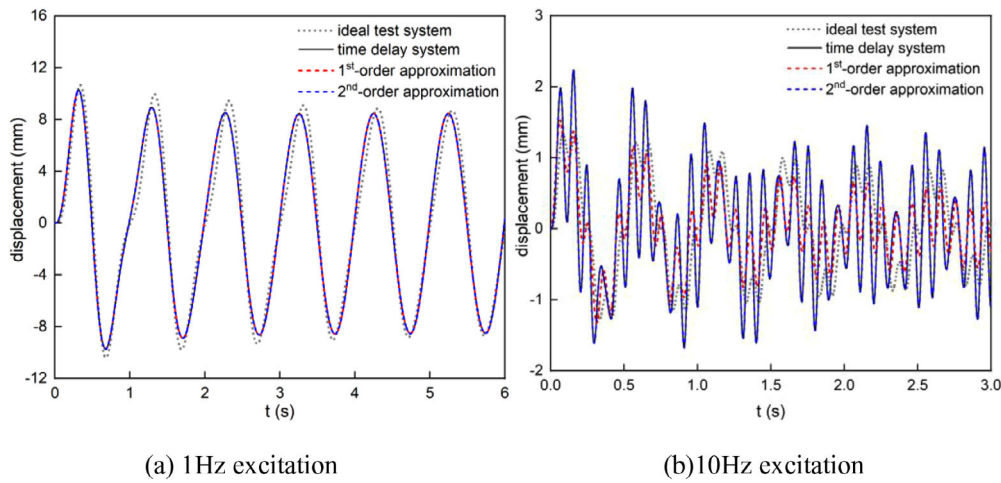


FIGURE 5 | Displacement response under a harmonic excitation for a time-delay system.

threshold for the first-, second-, and third-order approximation at frequencies below 60.5 rad/s (9.6 Hz), 165 rad/s (26.3 Hz), and 304 rad/s (48.4 Hz).

Figure 5 further illustrates the energy distribution across each mode. For example, at an external force frequency of $\omega = 2\pi$ rad/s (1 Hz), the time-delay system absorbs 0.66% more energy than the ideal system. The energy distribution in the time-delay system is 99.6% in the first-mode, 0.035% in the second-mode, and less than 0.01% in higher modes, indicating the dominant role of the first mode. At $\omega = 20\pi$ rad/s (10 Hz), the time-delay system absorbs 219.46% more energy, with the first-, second-, and third- modes contributing 50.54%, 49.17%, and less than 0.25%, respectively, suggesting that the first and second modes have nearly equal significance. According to Equations (18a) and (18b), we acknowledge that the test results under 1 Hz excitation are reliable, while those under 10 Hz excitation are unreliable.

4.2 | Response to General Dynamic Loading

The above analysis demonstrates that the dynamic response of time delay system is influenced by both internal and

external factors. The former includes the system's structural properties ($\mathbf{M}, \mathbf{C}, \mathbf{K}$), substructure partition ($\mathbf{M}^N : \mathbf{M}^E, \mathbf{C}^N : \mathbf{C}^E, \mathbf{K}^N : \mathbf{K}^E$), and time-delay characteristics (τ). The latter is determined by the frequency characteristic of the excitation. When the time-delay system is subjected to general dynamic loading, the response is obtained through load decomposition and response superposition. First, the Fourier transform decomposes the general dynamic loading into harmonic components; next, the dynamic response to each harmonic load is evaluated; and last, the total response is obtained by superposition.

For a continuous-time signal, the Fourier transform decompose an arbitrary dynamic loading $p(t)$ into an infinite sum of harmonic components. For a periodic dynamic loading with period of T_p , it can be expressed as

$$p(t) = \sum_{n=-\infty}^{+\infty} P_n \exp(i\omega_n t) \quad (19a)$$

where $\omega_n = \frac{2\pi n}{T_p}$, and $P_n = \frac{1}{T_p} \int_0^{T_p} p(t) \exp(-i\omega_n t) dt$. Similarly, for a discrete-time signal, the discrete Fourier transform (DFT) and

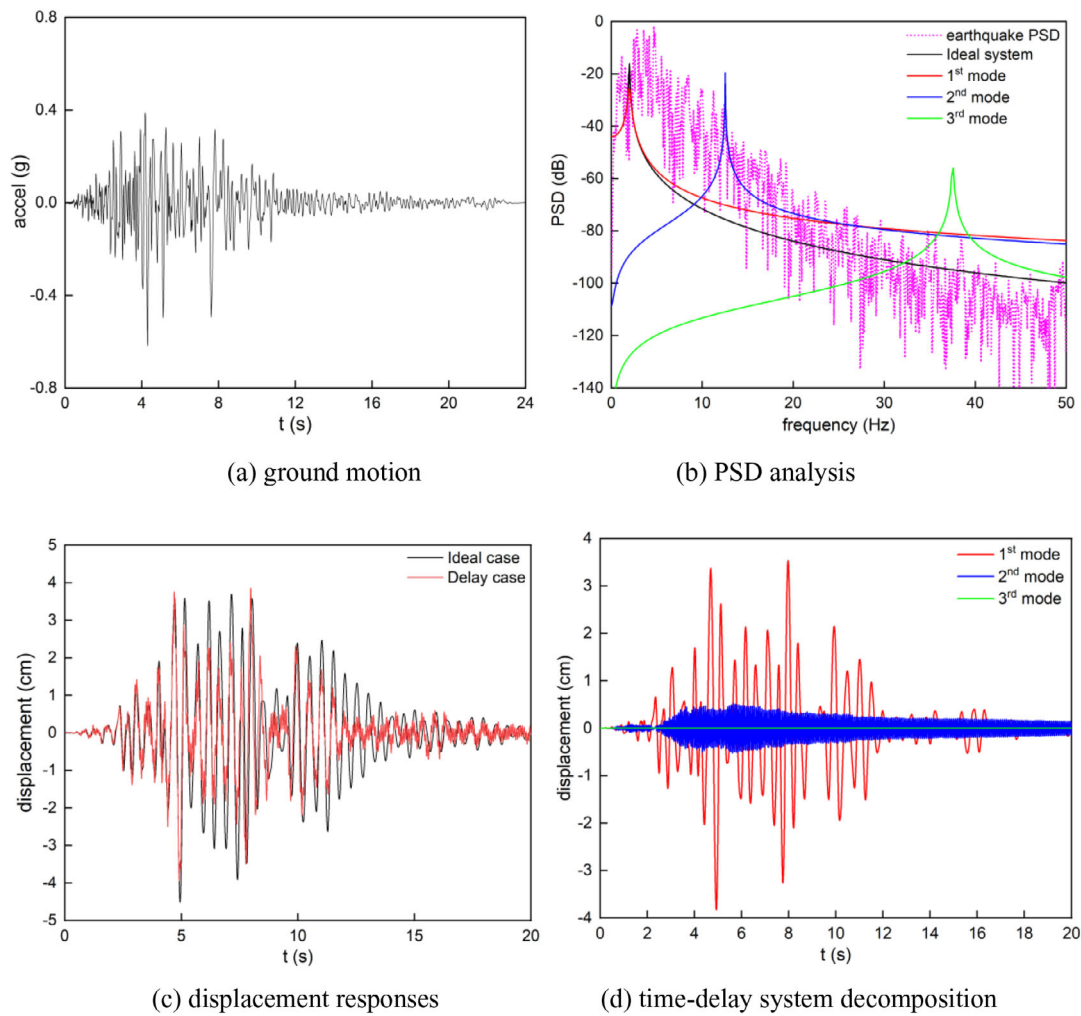


FIGURE 6 | Response analysis of time delay system subject to Northridge wave.

fast Fourier transform (FFT) also can be applied to decompose an arbitrary dynamic loading into a sum of an infinite number of harmonic loadings. Substituting Equation (19a) into the transfer function in Equation (7b) or Equation (12c) leads to the dynamic response under a general dynamic loading.

$$x(t) = \sum_{n=-\infty}^{+\infty} H(i\omega_n) P_n \exp(i\omega_n t) \quad (19b)$$

To verify the above derivation, the seismic ground motion from the Northridge (1994) earthquake is used for computational analysis, with the external force $f(t) = m\ddot{u}_g(t)$ representing the earthquake force. The ground motion record and its power spectral density (PSD) analysis are shown in Figure 6a,b, respectively. These figures illustrate the time-domain characteristics of the seismic record and the frequency-domain distribution of energy across different frequencies, allowing for a detailed assessment of the input excitation's spectral properties.

Figure 6b indicates that the energy of earthquake excitation is concentrated between 3~5 Hz, which corresponds to the frequency range between the first and second modes of the time-delay system. Figure 6c compares the displacement responses of the ideal system (black line) and the time-delay system

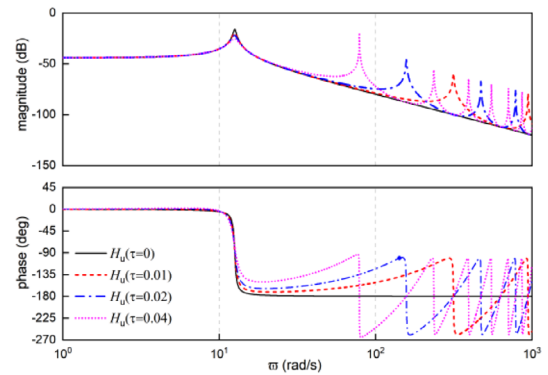


FIGURE 7 | Frequency response function with different time delays.

(red line). The displacement response of the time-delay system exhibits more high-frequency vibrations compared to the ideal system. Figure 6d decomposes the displacement response of the time-delay system into the first three modes. The first-, second-, and third-order approximate displacement responses are represented by red, blue, and green lines, respectively. According to Equations (17a)–(17b), it is determined that in the ideal system receives 0.2610 J of energy from the external force, whereas the

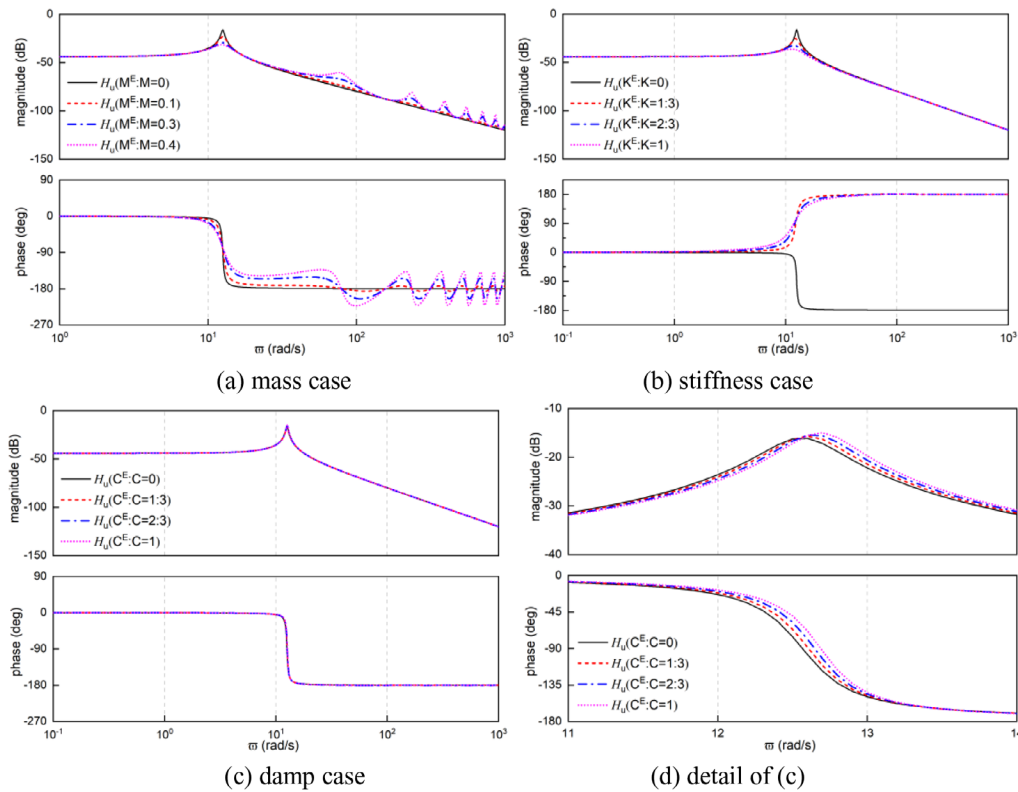


FIGURE 8 | Frequency response function with different substructure division.

time delay system receives a total of 1.0331 J. Specifically, the first-, second-, and $\geq$ third- modes receive 0.9056 J, 0.1278 J, and less than 0.001 J, respectively, accounting for 87.58%, 12.36% and 0.06% of the total energy input. These results indicate that the time delay causes higher energy input, which is consistent with previous findings.

4.3 | Parametric Study

From the above analysis, it is evident that actuator delay alters the dynamic characteristics of RTHS system. There are three primary strategies to control experimental errors: (1) reduce actuator delay, (2) reduce the experimental substructure ratio, and (3) coordinate the spectra between external excitation and the system's dynamic response.

In RTHS, the most direct and effective method to mitigate the experiment error is delay compensation. As illustrated in the case above, when the time delay of the test system varies from 0 to 0.04 s, the corresponding frequency response functions are shown in Figure 7. As the time delay decreases, the spurious modes move further away from the inherent modes. Particularly, when the time delay approaches zero $\tau \rightarrow 0_+$, the second-mode tends to infinity, and the system response approximates that of the ideal case.

Alternatively, reducing the experimental substructure ratio is another effective way to control testing error. For a prototype structure with the time delay of $\tau = 0.04$ s, Table 2 shows the parameters of cases with varying ratios of mass, damping, and stiffness.

TABLE 2 | Parameters of substructure division.

Substructure division	M^E/M	C^E/C	K^E/K
Mass case	$0, \frac{1}{10}, \frac{3}{10}, \frac{4}{10}$	0	0
Damping case	0	$0, \frac{1}{3}, \frac{2}{3}, 1$	0
Stiffness case	0	0	$0, \frac{1}{3}, \frac{2}{3}, 1$

Figure 8 compares the frequency response functions for different substructure ratios, as shown in Table 2. Figure 8a demonstrates that reducing the mass ratio of the experimental substructure $\frac{M^E}{M}$ diminishes the peaks of spurious higher modes, indicating that decreasing the mass ratio helps control error. Figure 8b shows that as the stiffness ratio of the experimental substructure $\frac{K^E}{K}$ decreases, the peak of the inherent model approaches the ideal case, suggesting that reducing the stiffness ratio also aids in error control. Figure 8c,d reveal that as the damping ratio $\frac{C^E}{C}$ of the experimental substructure increases, the peak of the inherent mode rises and shifts to the right, indicating that reducing the damping ratio is beneficial for controlling test errors.

Last, coordinating the spectrum between external excitation and the system's dynamic response can also effectively control test error. As shown in Figure 5, for the same RTHS system and the same time delay, varying the excitation frequency from 1 to 10 Hz significantly affects the energy distribution in the inherent mode. The energy in the inherent mode decreases from 99.6% to 50.62%, demonstrating that aligning the excitation

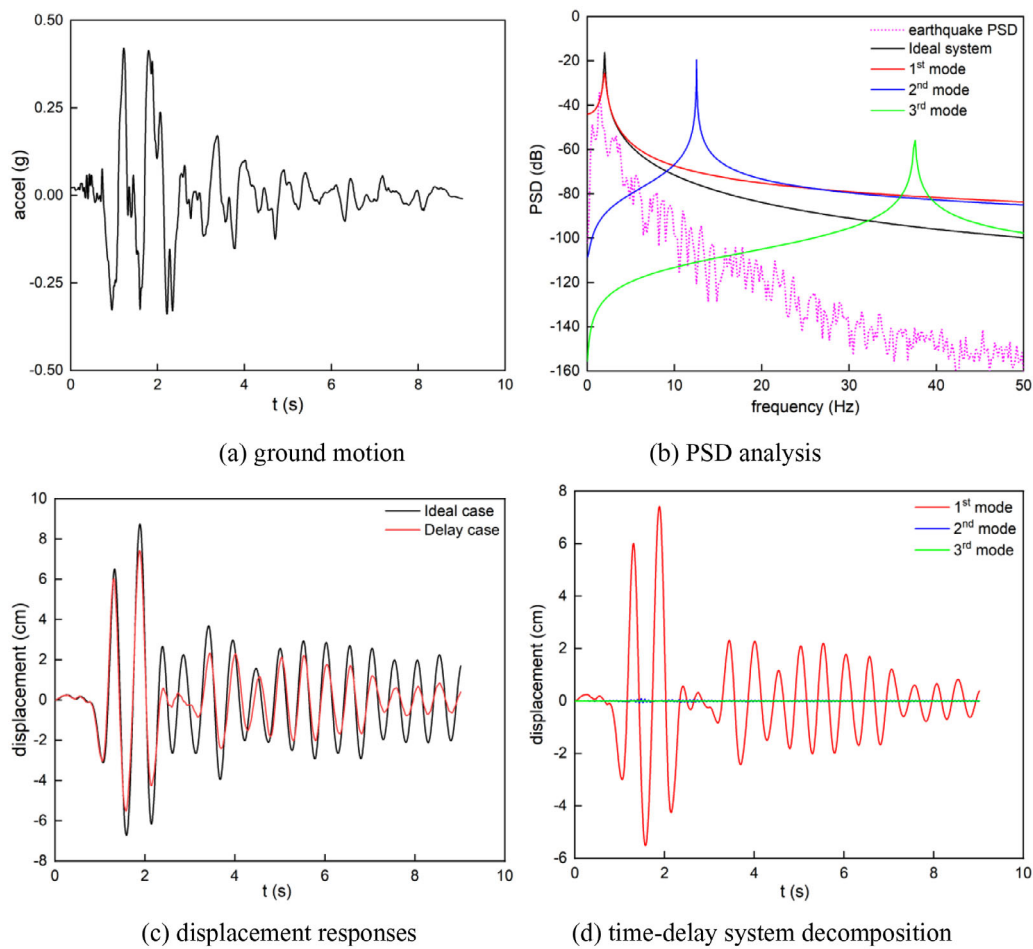


FIGURE 9 | Response analysis of time delay system subject to San Salvador wave.

TABLE 3 | Harmonic and ground excitation.

Type	Name	Test	Time (s)	Freq (rad/s)	$\ddot{u}_g$ (gal)
Harmonic	ϖ_1	Exp 101–105	60	0.2π	8
	ϖ_2	Exp 201–205	60	0.5π	6
	ϖ_3	Exp 301–305	60	2π	6
	ϖ_4	Exp 401–405	60	4π	10
Earthquake	El-Centro	EL 001–005	60	—	10
	Taft	TF 001–005	60	—	10



FIGURE 10 | Experimental test setup.

spectrum with the system response can significantly reduce test errors.

A similar phenomenon is observed for ground motions. For instance, consider the San Salvador (1986) seismic wave, where the numerical and experimental substructures and actuator delay are the same as Figure 6. However, Figure 9 illustrates notable differences compared to Figure 6. Figure 9a,b shows the acceleration and PSD of the ground motion, respectively. Unlike the Northridge wave, the seismic energy of the San Salvador wave is concentrated in low-frequency bands. Figure 9c shows the displacement response of ideal and delay systems, which have inputs 0.3809 and 0.4484 J of energy, respectively. Figure 9d decomposes the input energy into the first three modes: 0.4415

TABLE 4 | Test error and eigenvalue analysis.

Type	α	τ (ms)	$\lambda_1, \bar{\lambda}_1$	$\lambda_2, \bar{\lambda}_2$	$\lambda_3, \bar{\lambda}_3$
Ideal	1	0	$-0.1571 \pm 3.1378i$	/	/
Exp101	1.0043	5.1002	$-0.1545 \pm 3.1386i$	$-3172.6 \pm 701.29i$	$-3238.6 \pm 2071.0i$
Exp102	1.0044	5.2752	$-0.1545 \pm 3.1386i$	$-3052.9 \pm 678.45i$	$-3117.2 \pm 2003.2i$
Exp103	1.0044	5.3274	$-0.1544 \pm 3.1386i$	$-3018.8 \pm 671.93i$	$-3082.6 \pm 1983.8i$
Exp104	1.0044	5.4983	$-0.1544 \pm 3.1386i$	$-2911.9 \pm 651.44i$	$-2974.3 \pm 1922.9i$
Exp105	1.0044	5.5565	$-0.1543 \pm 3.1386i$	$-2877.1 \pm 644.74i$	$-2939.0 \pm 1903.0i$
Exp201	1.0101	11.3170	$-0.1514 \pm 3.1395i$	$-1268.6 \pm 321.46i$	$-1305.2 \pm 943.41i$
Exp202	1.0101	11.3673	$-0.1514 \pm 3.1395i$	$-1262.1 \pm 320.08i$	$-1298.6 \pm 939.30i$
Exp203	1.0101	11.3796	$-0.1514 \pm 3.1395i$	$-1260.5 \pm 319.74i$	$-1297.0 \pm 938.30i$
Exp204	1.0097	11.4540	$-0.1514 \pm 3.1395i$	$-1251.0 \pm 317.71i$	$-1287.4 \pm 932.29i$
Exp205	1.0097	11.4107	$-0.1514 \pm 3.1395i$	$-1256.5 \pm 318.89i$	$-1293.0 \pm 935.77i$
Exp301	1.0192	14.7310	$-0.1497 \pm 3.1410i$	$-932.62 \pm 248.64i$	$-962.98 \pm 727.66i$
Exp302	1.0189	14.6887	$-0.1497 \pm 3.1409i$	$-935.78 \pm 249.34i$	$-966.20 \pm 729.72i$
Exp303	1.0188	14.6677	$-0.1497 \pm 3.1409i$	$-937.35 \pm 249.69i$	$-967.80 \pm 730.75i$
Exp304	1.0176	14.6788	$-0.1497 \pm 3.1409i$	$-937.35 \pm 249.69i$	$-967.80 \pm 730.75i$
Exp305	1.0176	14.6705	$-0.1497 \pm 3.1407i$	$-937.24 \pm 249.64i$	$-967.68 \pm 730.60i$
Exp401	1.0244	15.8503	$-0.1490 \pm 3.1418i$	$-855.70 \pm 231.56i$	$-884.53 \pm 677.06i$
Exp402	1.0246	15.8146	$-0.1491 \pm 3.1418i$	$-857.95 \pm 232.06i$	$-886.82 \pm 678.56i$
Exp403	1.0245	15.8009	$-0.1491 \pm 3.1418i$	$-858.83 \pm 232.26i$	$-887.72 \pm 679.14i$
Exp404	1.0241	15.8999	$-0.1490 \pm 3.1417i$	$-852.60 \pm 230.85i$	$-881.36 \pm 674.98i$
Exp405	1.0235	15.8359	$-0.1491 \pm 3.1417i$	$-856.68 \pm 231.76i$	$-885.52 \pm 677.66i$
EL001	1.0151	13.5101	$-0.1503 \pm 3.1403i$	$-1032.03 \pm 270.49i$	$-1064.32 \pm 792.36i$
EL002	1.0150	13.4713	$-0.1503 \pm 3.1403i$	$-1035.50 \pm 271.25i$	$-1067.86 \pm 794.60i$
EL003	1.0149	13.4970	$-0.1503 \pm 3.1403i$	$-1033.21 \pm 270.74i$	$-1065.52 \pm 793.11i$
EL004	1.0150	13.4975	$-0.1503 \pm 3.1403i$	$-1033.16 \pm 270.73i$	$-1065.47 \pm 793.09i$
EL005	1.0150	13.5166	$-0.1503 \pm 3.1403i$	$-1031.46 \pm 270.36i$	$-1063.74 \pm 791.98i$
TF001	1.0141	13.4463	$-0.1503 \pm 3.1402i$	$-1037.82 \pm 271.74i$	$-1070.22 \pm 796.05i$
TF002	1.0141	13.4168	$-0.1504 \pm 3.1402i$	$-1040.48 \pm 272.32i$	$-1072.93 \pm 797.78i$
TF003	1.0140	13.3906	$-0.1504 \pm 3.1402i$	$-1042.86 \pm 272.84i$	$-1075.35 \pm 799.32i$
TF004	1.0140	13.3543	$-0.1504 \pm 3.1402i$	$-1046.16 \pm 273.56i$	$-1078.72 \pm 801.46i$
TF005	1.0141	13.3342	$-0.1504 \pm 3.1402i$	$-1047.99 \pm 273.96i$	$-1080.59 \pm 802.65i$

J for the first mode, 0.0067 J for the second mode, and less than 0.001 J for the third mode, accounting for 98.45%, 1.49%, and 0.02% of the total energy input. It is evident that since the seismic energy input is more concentrated in the structural inherent mode and less in higher-order spurious modes, the impact of time delay is reduced. Comparing Figure 9 to Figure 6, it is found that the delay effect of RTHS is the result of the combined action of the dynamic characteristics of the testing system (internal factors) and external excitation (external factors), different seismic excitations can cause substantial distortion in test results or yield acceptable test errors for the same test system.

5 | Experimental Study

5.1 | Test Setup

A series of RTHS tests were conducted in the Key Laboratory of Concrete and Prestressed Concrete Structures at Southeast University to experimentally validate the findings. The tests utilized real-time multi-directional earthquake simulation equipment by MTS Corporation. This system consists of an 8-channel digital controller with a 1024 Hz clock speed, which controls the actuators through a closed servo-control loop. The software contains the OpenSees, OpenFresco, Simulink, MTS793, etc.

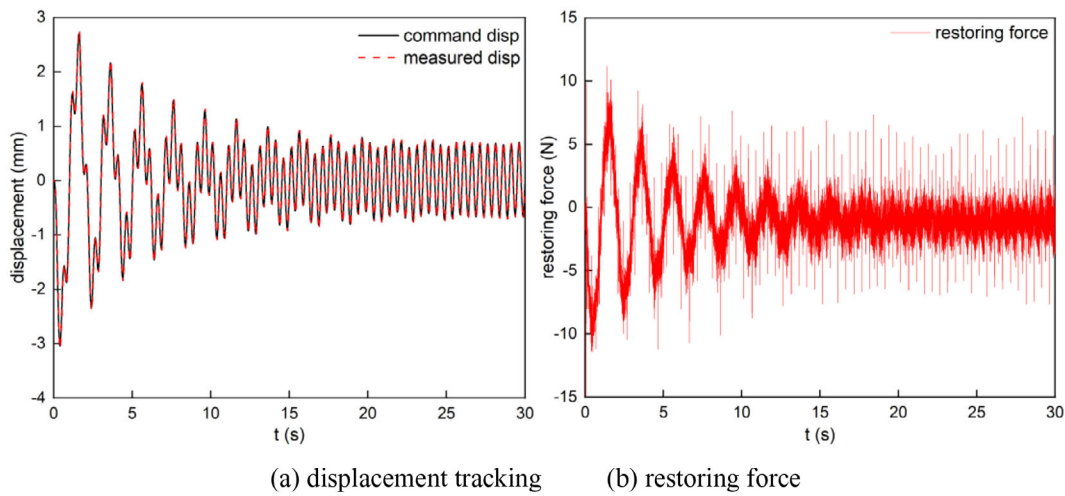


FIGURE 11 | Displacement and restoring force responses of Exp404.

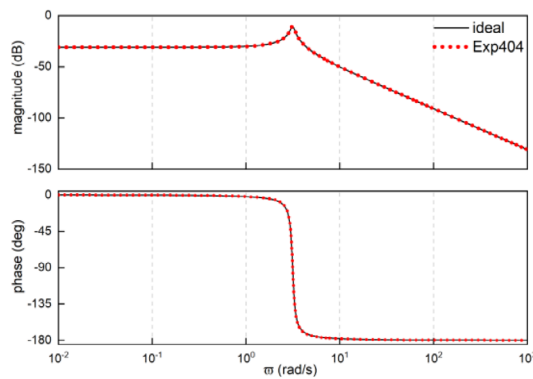


FIGURE 12 | Bode plots of Exp 404 and ideal case.

The test flow of RTHS is run in Simulink; the numerical substructure is built in OpenSees; the experimental substructure is pushed by a servo-hydraulic actuator 244.21 through MTS793. Communication and data coupling between computational analysis and experimental loading are facilitated by OpenFresco.

For the experiments, the numerical substructure consists of a one-story one-bay frame with a single horizontal DOF, while the experimental substructure is an ideal spring. The parameters of the substructures are $M^N = 3546$ kg, $C^N = 1.114$ N.s/mm, $K^N = 31.5$ N/mm, $M^E = 0$ kg, $C^E = 0$ N.s/mm, and $K^E = 3.5$ N/mm. The experimental test setup is shown in Figure 10. As shown in Table 3, four different harmonic waves $\varpi_1 \sim \varpi_4$ and two earthquake records, EL-Centro (1940) and Taft (1952), are selected as external excitations. Both ground motion records have a sampling frequency of 50 Hz. Each test continues for 60 s after the excitation input ends, and each excitation is repeated five times to ensure repeatability. The generalized- α algorithm [46] is used for numerical integration with the integration time step of 20/1024 s, while the sampling time of servo-hydraulic system is 1/1024 s. To better understand the effects of time delay, no compensation or correction method is applied in any of the experiments. The initial condition of the structure is static.

5.2 | Experimental Results

After the tests, the frequency-domain evaluation index (FEI) [14] was applied to analyze the average amplitude ratio and time delay, as summarized in Table 4. Utilizing the FFT, the FEI decomposes the command and measured displacement signals of the entire test duration in the frequency domain, providing the amplitude ratio (α) and time delay (τ) across all frequencies as the evaluation indices. The results indicate that both amplitude ratio error and time delay increase as the frequency of sinusoidal waves rises.

5.3 | Modal Analysis

Table 4 further presents the first-, second-, and third-mode of the time-delay system. Among the 25 tests, Exp 404 exhibits the highest amplitude ratio error and time delay, resulting in the smallest distance between adjacent poles. Figure 11 shows the displacement responses of Exp 404. It can be observed in Figure 11a that the servo-hydraulic actuator dynamics introduce a noticeable time delay and amplitude error between the command and measured displacements. Figure 11b displays the measured restoring force, which contains some measurement noise.

Figure 12 compares the Bode plots of Exp 404 (red dots) with the ideal case (black line). It is evident that the magnitude and phase plots of the time-delay system closely resemble those of corresponding idea case. For the ideal case, the magnitude plot exhibits a single peak at $\Omega_1 = 3.1378$ rad/s (0.499 Hz). Due to time delay, the RTHS system introduces spurious high-order modes. However, the adjacent second-mode at $\Omega_2 = 230.85$ rad/s (36.74 Hz) is sufficiently distant from the first-mode at $\Omega_1 = 3.1417$ rad/s (0.500 Hz). Additionally, the magnitude of the RTHS system $|H(i\Omega_2)| = 5.2926 \times 10^{-6}$ is nearly identical to that of ideal case $|H(i\Omega_2)| = 5.2928 \times 10^{-6}$. Therefore, the actuator delay and amplitude error have minimum adverse effect on the RTHS system, and the influence of spurious high-modes can be ignored, with the first-mode remaining dominant.

Furthermore, for Exp 404, the frequency of the external force is $\varpi_4 = 4\pi$ rad/s. Figure 12 indicates that for the ideal

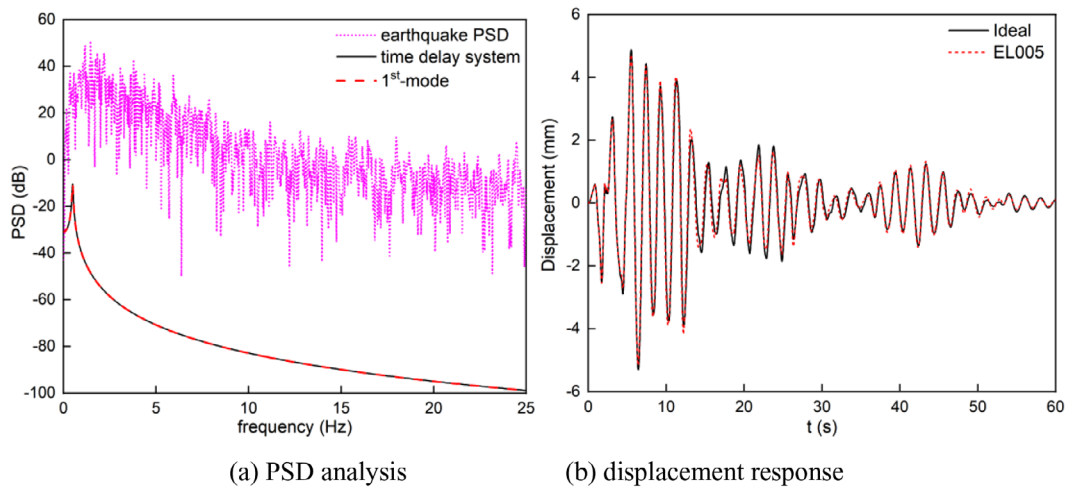


FIGURE 13 | Spectral analysis and displacement response of EL005.

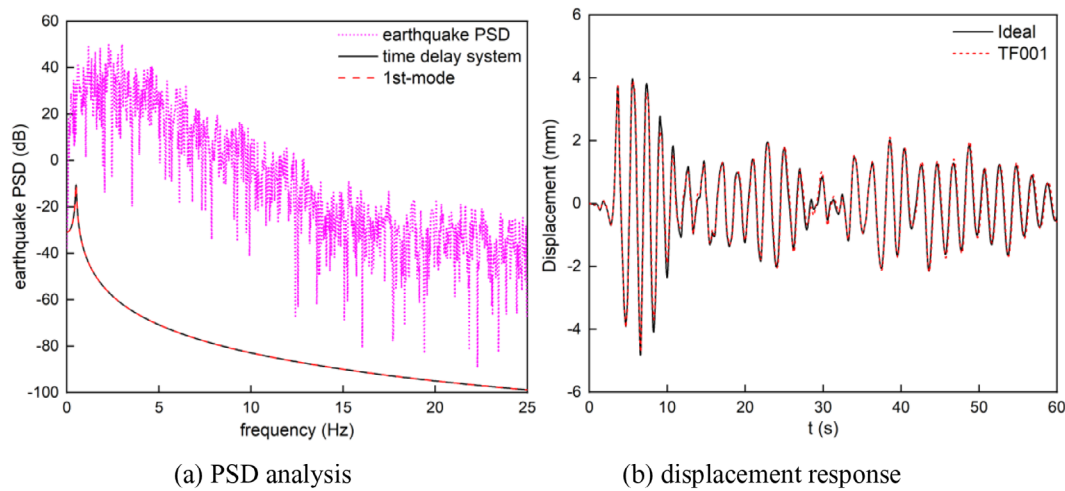


FIGURE 14 | Spectral analysis and displacement response of TF001.

system $|H(i\varpi_4)| = 1.9042 \times 10^{-3}$ and $\angle H(i\varpi_4) = -178.472^\circ$ remain consistent, while for the time-delay system, $|H(i\varpi_4)| = 1.9043 \times 10^{-3}$ and $\angle H(i\varpi_4) = -178.550^\circ$ show a marginal increase. This implies that the time delay adds only 0.006% additional energy to the system and nearly 99.99% energy input is concentrated in the first-mode, confirming that the test results are reliable.

5.4 | Earthquake Analysis

When a time delay system is subjected to general dynamic loading, load decomposition and response superposition can be applied to evaluate the impact of time delay on RTHS. Table 4 indicates that the tests of EL005 and TF001 have the largest loading error, making them the focus of further analysis. Figures 13a and 14a present the PSD of El-Centro and Taft ground motions, respectively, showing that the energy is concentrated within 0–25 Hz frequency range for both two ground motions. Affected by actuator delay, the RTHS system introduces 2.62% more energy under El-Centro ground motion and –2.85% less energy under

Taft ground motion. Moreover, over 99.99% of energy input is concentrated in the first-mode, while the contributions from the second- and higher-order modes are negligible, since the spurious modes are located far away from the first-mode. Consequently, the small amplitude ratio error and time delay have a minimum adverse effect on the accuracy of RTHS.

Figures 13b and 14b compare the time-domain displacement responses with the ideal case. Despite the presence of amplitude error and time delay, the test results remain highly consistent with the ideal case, confirming the negligible influence of time delay in these scenarios.

6 | Summary and Conclusion

RTHS inherently functions as a feedback-controlled system with time-delay. Time delay not only alters the inherent modes of structure but also introduces spurious modes, which are the primary cause of system instability. Existing research inadequately quantifies the impact of time delay on structural vibration. To

address this gap, this study introduces a novel spectral decomposition method based on the theory of functional differential equations. Through three steps of spectral decomposition, modal projection, and equivalence substitution, this approach simplifies a linear time-delay system with infinite modes into a finite set of single-mode systems, which can effectively quantify the frequency excursions and spurious modes induced by time delay, and analyze the energy redistribution across inherent and spurious modes.

The study highlights three primary strategies to mitigate test errors: (1) reducing actuator delay, (2) minimizing the experimental substructure ratio, and (3) aligning the spectral characteristics between the external excitation and system response. Additionally, it suggests two energy evaluation indexes and their tolerances, reflecting the variation of energy input caused by time delay and the concentration of energy input in the main mode, respectively. Both computational simulations and laboratory experiments validate the effectiveness of the spectral decomposition method, demonstrating its capability to accurately predict the dynamic behavior of RTHS.

This research advances the understanding of time-delay effects by providing a robust analytical framework to evaluate errors and guide error control strategies, offering a significant innovation in enhancing the reliability and accuracy of RTHS outcomes. While the current spectral decomposition framework does not incorporate active delay compensation mechanisms, it offers detailed insight into the modal effects of time delay—particularly the amplification of spurious modes and redistribution of energy. These insights can inform the design of delay compensation strategies by identifying critical modes most susceptible to destabilization. Integrating compensation techniques, such as model-based predictors or adaptive control, with the proposed spectral analysis represents a promising direction for future research to enhance the stability and accuracy of RTHS. It should be noted, however, that the present study is limited to linear elastic systems. Future research will extend the approach to accommodate nonlinear structural behavior.

Acknowledgments

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix

Suppose that Λ is a finite set $\{\lambda_1 \dots \lambda_p\}$ of eigenvalues of Equation (7a) or (9), λ_j , $\bar{\lambda}_j = \alpha_j \pm \beta_j i$. Let $\Phi_\Lambda = (\Phi_{\lambda_1} \dots \Phi_{\lambda_p})$, $A_\Lambda = \text{diag}(A_{\lambda_1} \dots A_{\lambda_p})$, $A_{\lambda_j} = \begin{bmatrix} \alpha_j & \beta_j \\ -\beta_j & \alpha_j \end{bmatrix}$, where Φ_{λ_j} is a basis for the generalized eigenspace of λ_j .

$$A \Phi_\Lambda = \Phi_\Lambda A_\Lambda \quad (\text{A1})$$

$$\Phi_\Lambda(\theta) = \Phi_\Lambda(0) \exp(A_\Lambda \theta), -\tau \leq \theta \leq 0 \quad (\text{A2})$$

Design the adjoint system as Equation (A3), which has the same eigenvalues to Equation (7a).

$$\dot{\mathbf{x}}(t) - \dot{\mathbf{x}}(t + \tau) A_2 = -\mathbf{x}(t) A_0 - \mathbf{x}(t + \tau) A_1 \quad (\text{A3})$$

Similarly, A^* is the formal adjoint of A , Ψ^* is a basis for the generalized eigenspace.

$$\Psi^* A^* = A_\Lambda \Psi^* \quad (\text{A4})$$

$$\Psi^*(\xi) = \exp(A_\Lambda \xi) \Psi^*(0), 0 \leq \xi \leq \tau \quad (\text{A5})$$

Orthogonalize Ψ^* and Φ_Λ to satisfy $\langle \Psi_\Lambda, \Phi_\Lambda \rangle = I$.

$$\Psi_\Lambda(\xi) = \langle \Psi^*, \Phi_\Lambda \rangle^{-1} \Psi^*(\xi) \quad (\text{A6})$$

$$\begin{aligned} \langle \psi, \phi \rangle &= \psi(0) \phi(0) + \int_{-\tau}^0 \psi(\xi + \tau) A_1 \phi(\xi) d\xi \\ &+ \left[\frac{d}{d\theta} \int_{\theta}^0 \psi(\xi - \theta) A_2 \phi(\xi) d\xi \right]_{\theta=-\tau} \quad (\text{A7}) \end{aligned}$$